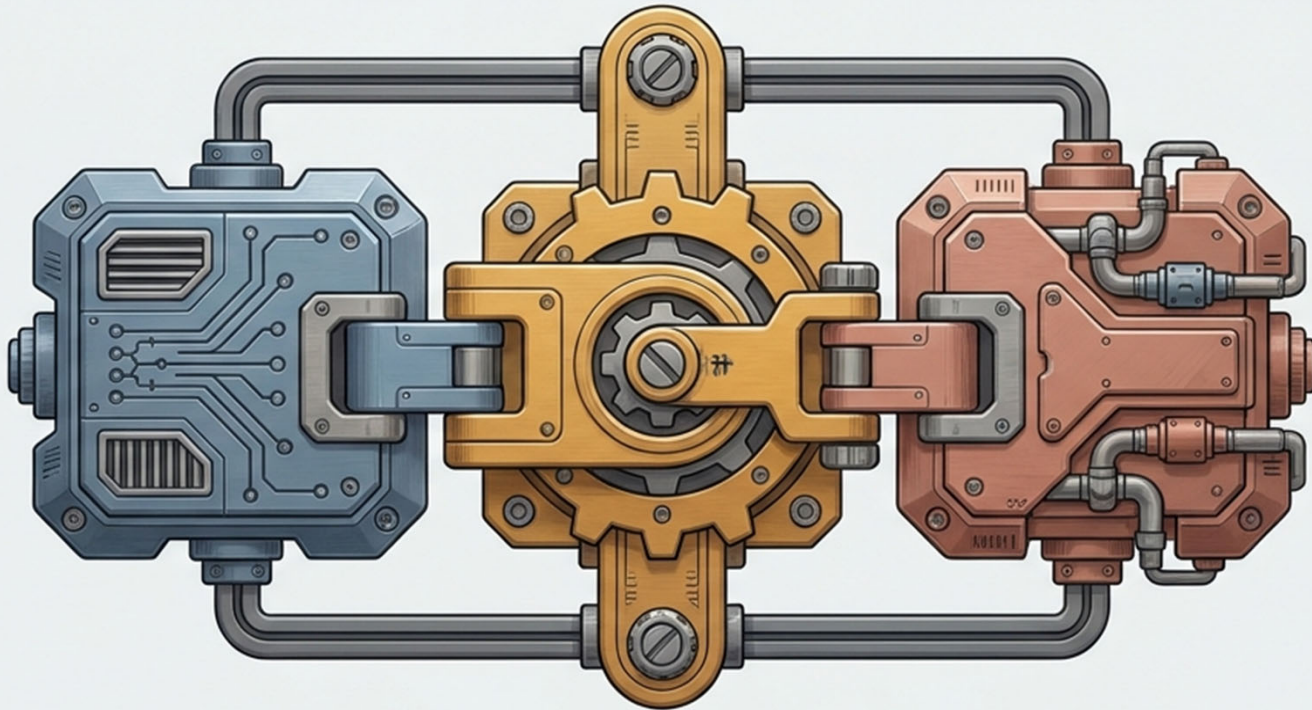


Visual Instruction Tuning



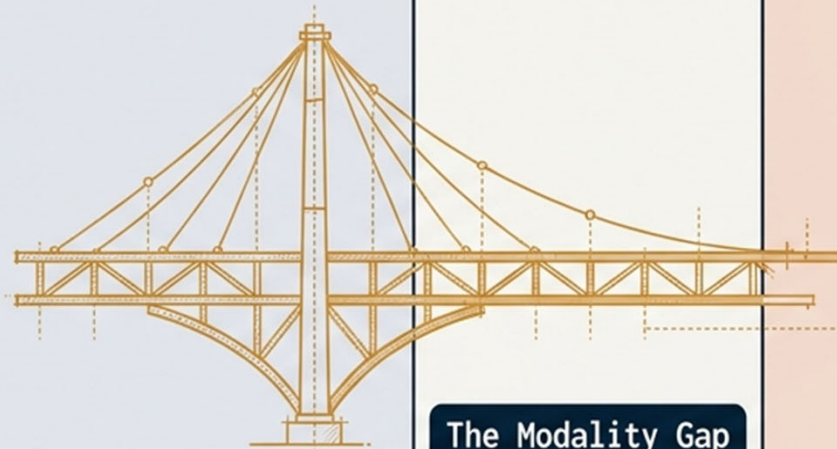
From Passive Chatbots to
Omni-Modal Operating Systems

Vision Models

[Classification]

[Detection]

[Segmentation]



The Modality Gap

Understands pixels, but cannot follow conversational instructions.

Large Language Models

[Reasoning]

[Generation]

[Zero-shot]



Understands complex instructions, but cannot see the physical world.

Visual Instruction Tuning (VIT) is the engineering practice of building the bridge. It explicitly trains models to map visual signals to language semantics, creating a unified general-purpose assistant.

01

The Communicator (LLaVA)

Focus: Data & Concept

Solved multimodal data scarcity by using GPT-4 to generate instruction-following data.

02

The Efficiency Engine (BLIP-2)

Focus: Compute & Alignment

Solved catastrophic forgetting and pre-training costs using the Q-Former bottleneck between frozen models.

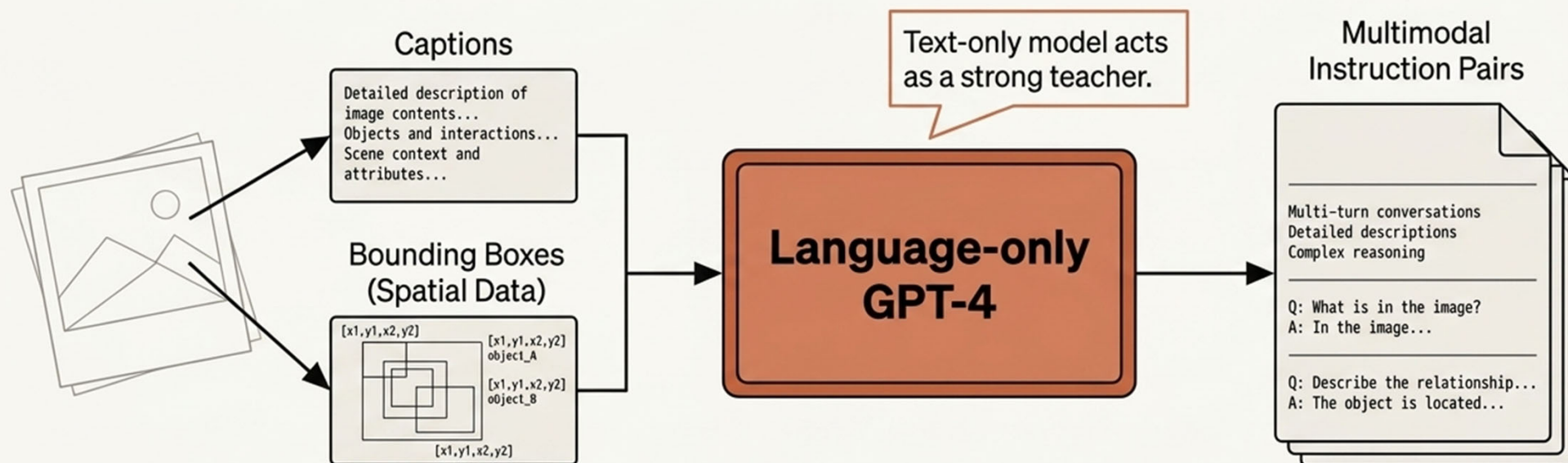
03

The Generalist (VisionLLM v2)

Focus: Action & Output

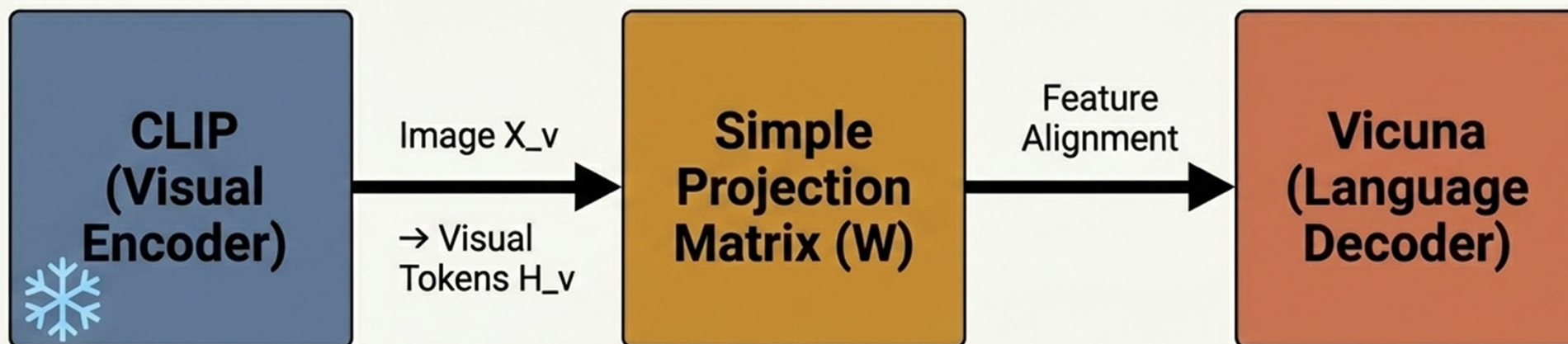
Broke the 'text-only' barrier using Super Links to output physical actions like segmentation and image generation.

The LLaVA Bootstrap: Synthetic Data Generation



Because human-annotated vision-language instruction data didn't exist at scale, LLaVA bypassed the bottleneck by feeding symbolic visual data (boxes and captions) to text-only GPT-4, successfully bootstrapping 158K multimodal instructional pairs.

LLaVA: The Linear Modality Bridge

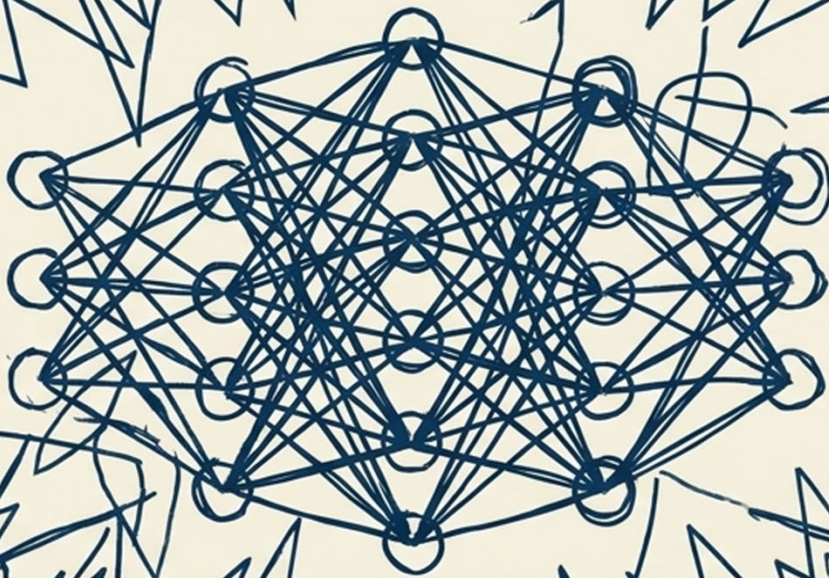


The Mechanism

LLaVA relies on a lightweight, linear projection matrix. During training, the visual encoder remains frozen, while the simple projection layer acts as a translator, allowing the LLM to read image features as if they were foreign language text tokens.

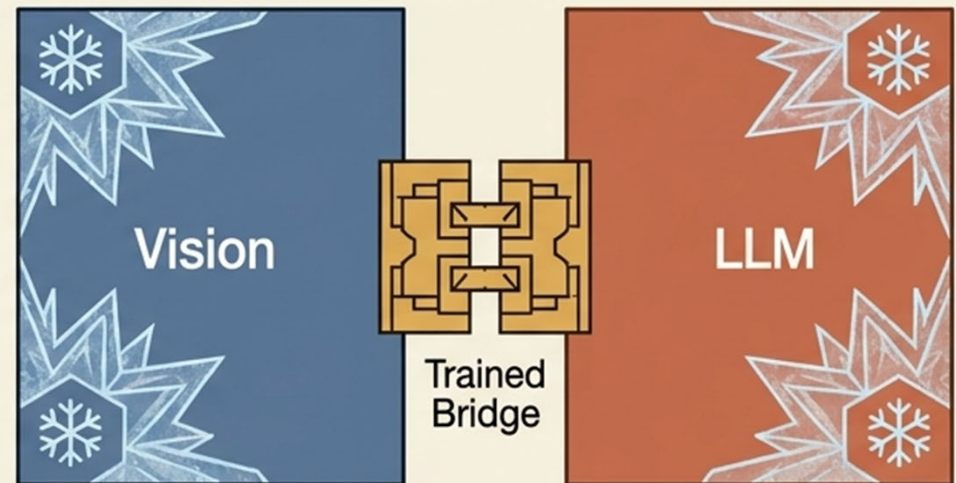
The BLIP-2 Imperative: Cost & Catastrophic Forgetting

The End-to-End Problem



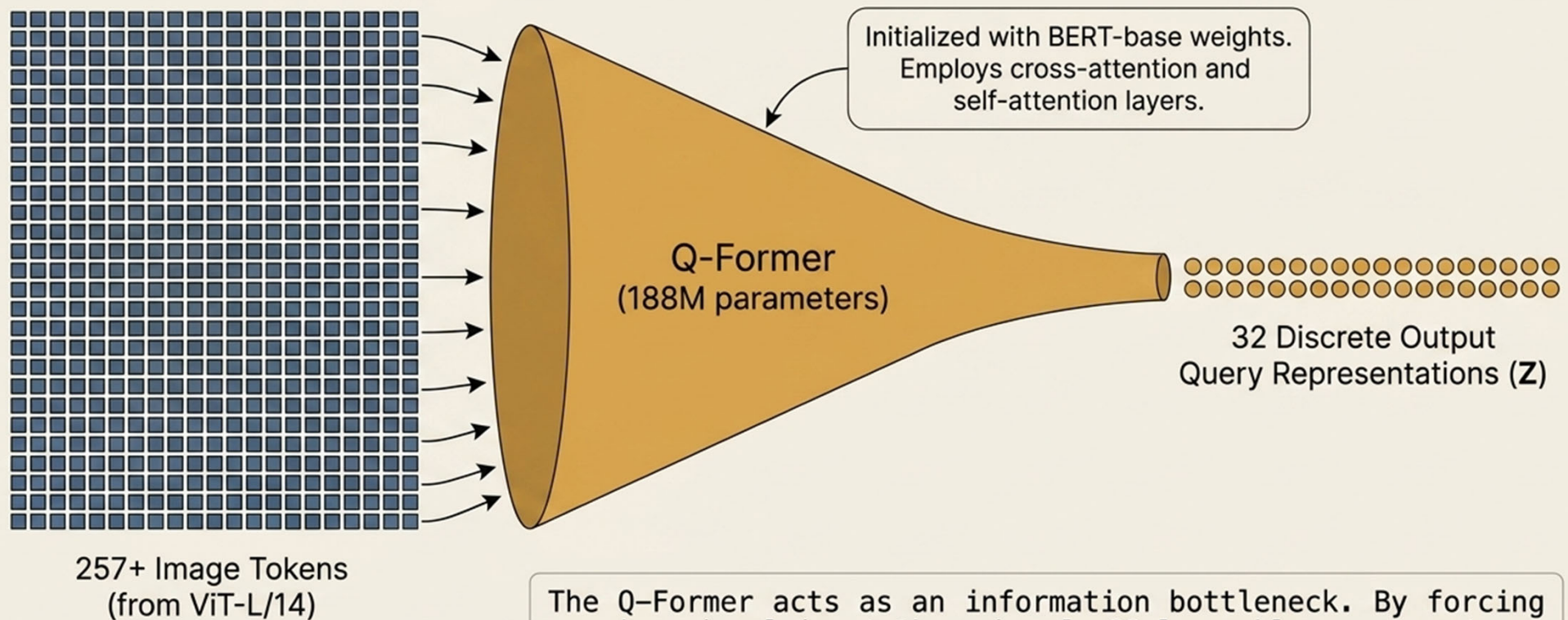
Prohibitively expensive compute costs.
Unfreezing LLM parameters risks “catastrophic forgetting” of existing text capabilities.

The Modular Solution



BLIP-2 hypothesizes that vision and language communities already have perfect unimodal models.
The goal: harvest these off-the-shelf models—keeping them completely frozen—and only train the bridge between them.

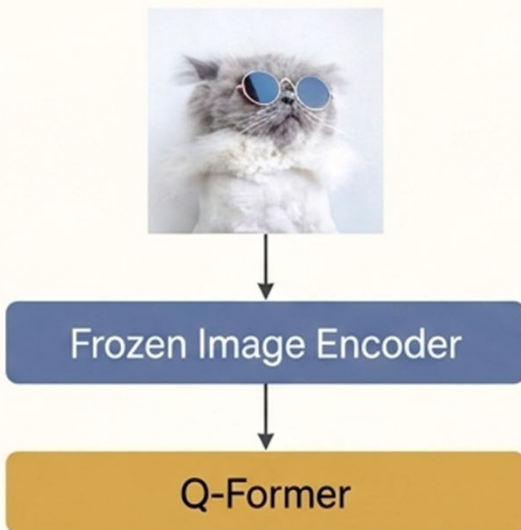
The Q-Former Information Bottleneck



The Q-Former acts as an information bottleneck. By forcing massive visual input through only 32 learnable query vectors, it explicitly filters out irrelevant visual noise and feeds the frozen LLM only the most language-informative features.

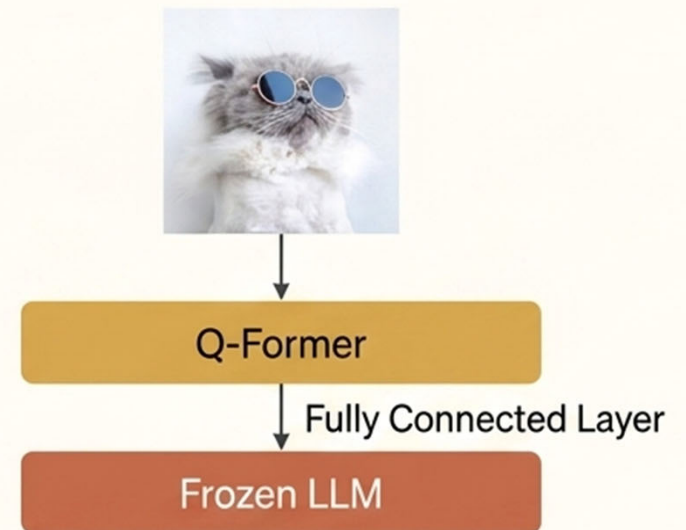
BLIP-2: Two-Stage Bootstrapping

Stage 1: Representation Learning



Forces the 32 queries to extract visual features relevant to text using three objectives: Image-Text Contrastive, Image-Grounded Text Gen, and Image-Text Matching.

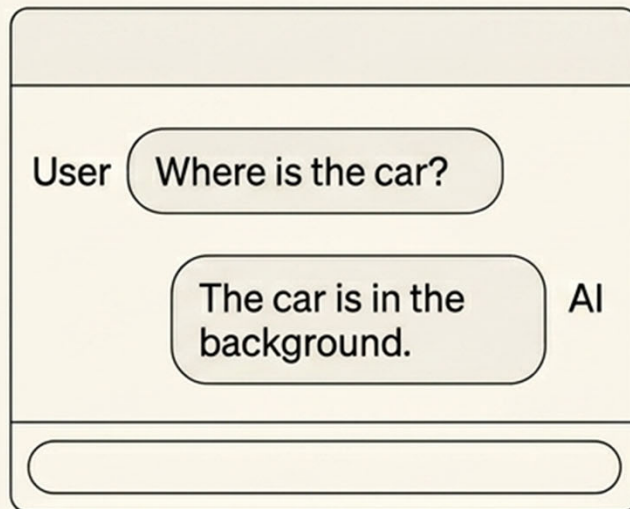
Stage 2: Generative Learning



The 32 extracted visual queries act as soft visual prompts. The frozen LLM learns to generate language conditioned on these compressed visual tokens.

VisionLLM v2: Breaking the Text-Only Barrier

The Limitation: Text-Only Models



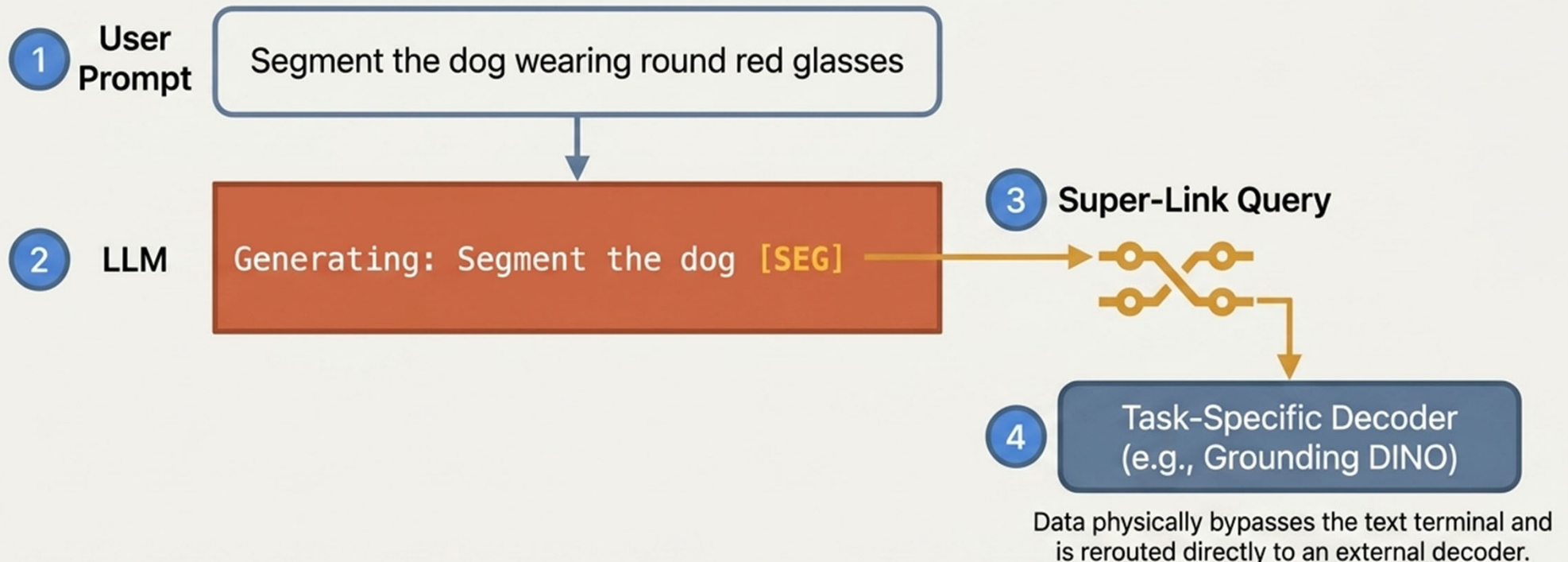
Traditional MLLMs can talk about an image, but they cannot draw a bounding box, mask an object, or generate a new image.

The Expansion: Generalist Omni-Modal Output



VisionLLM v2 unifies perception, understanding, and generation end-to-end, actively interacting with the physical and visual world.

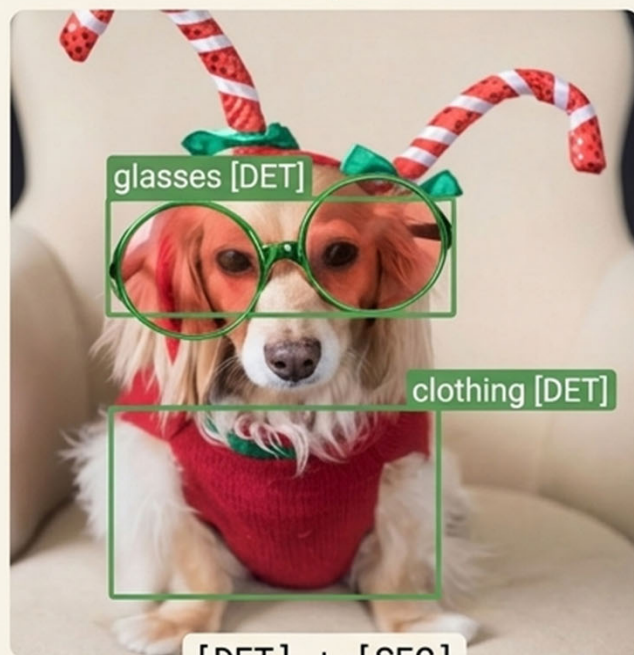
The Routing Mechanism: Super Links



Super Links are special tokens added to the LLM's vocabulary. When predicted, they instantly trigger randomly initialized learnable queries that extract task-specific information from the LLM and send it to an external decoder, resolving multi-task training conflicts.

Omni-Modal Outputs: An End-to-End Generalist

Perception



Simultaneous visual question answering and region detection.

Generation



Text-to-image generation via Stable Diffusion integration.

Editing



Instruction-based image editing and style transfer.

By utilizing open-ended decoders, users can freely define tasks through multimodal prompts—combining text, visual prompts, and in-context learning to interact with the world natively.

Architectures of Alignment: A Comparison

	LLaVA	BLIP-2	VisionLLM v2
Core Innovation	GPT-4 Data Generation	Frozen Unimodal Efficiency	Task Unification
Modality Bridge	Simple Linear Projection	Q-Former (Information Bottleneck)	Super Links (Routing Tokens)
Pre-Training Strategy	2-Stage End-to-End	2-Stage Bootstrap (Frozen)	3-Stage Joint Multi-Task
Output Capabilities	Text-only (Chat)	Text-only + Zero-Shot Instructed	Text + Detection + Segmentation + Poses + Image Gen

The Anatomy of an Omni-Modal Operating System

